CPIT 370- Project| **Phase-1** (2 Points)

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Your Network Address
192.100.100.0/24

Subnetting Table

Subnet Name	***n*** host bits	**Subnet mask** (in dot notation & CIDR)	**Subnet address** (Network address)	**First usable IP address**	**Last usable IP address**	**Broadcast address**
Branch2	6	255.255.255.192 /26	192.100.100.0	192.100.100.1	192.100.100.62	192.100.100.63
Branch1	5	255.255.255.224 /27	192.100.100.64	192.100.100.65	192.100.100.94	192.100.100.95
Branch3	4	255.255.255.240 /28	192.100.100.96	192.100.100.97	192.100.100.110	192.100.100.111
WAN	2	255.255.255.252 /30	192.100.100.112	192.100.100.113	192.100.100.114	192.100.100.115

Detailed Solution

In this project I was given the network 192.100.100.0/24 and asked to design several subnets using VLSM

The required subnets and their host requirements are:
Branch 2: 50 hosts
Branch 1: 20 hosts
Branch 3: 10 hosts
WAN link: 2 hosts

VLSM Design Strategy:

1. Sort the subnets from largest to smallest by required number of hosts.

2. For each subnet, calculate the number of host bits n , using the formula: 2^n - 2 >= required hosts
The value "−2" accounts for the network address and the broadcast address.

3. Compute the new prefix length: prefix = 32 – n

4. From the prefix , derive the subnet mask and the block size in the last octet:
block size = 256 - last octet of mask

5. Starting from the given network 192.100.100.0/24, assign each subnet a continuous block of addresses.
For each subnet :
-Subnet address
-First usable host address
-Last usable host address
-Broadcast address
-Next available address to start the following subnet

Step-by-Step Subnet Calculations:

The original network is 192.100.100.0/24, which provides 256 addresses (0–255) in the last octet.

Branch 2 (50 hosts):

1. Host bits:
2^6 - 2 = 62 >= 50 → n = 6

2. Prefix:
32 - 6 = 26 → /26

3. Subnet mask: 255.255.255.192
Block size = 256 − 192 = 64

4.allocate the first block (0–63):
Subnet address: 192.100.100.0
First usable host: 192.100.100.1
Last usable host: 192.100.100.62
Broadcast address: 192.100.100.63

The next free address after this subnet is 192.100.100.64.

Branch-1 (20 hosts):

1. Host bits:
2^5 - 2 = 30 >= 20 → n = 5

2. Prefix:
32 - 5 = 27 → /27

3. Subnet mask: 255.255.255.224
Block size = 256 − 224 = 32

4. Starting from 192.100.100.64:
Subnet address: 192.100.100.64
First usable host: 192.100.100.65
Last usable host: 192.100.100.94
Broadcast address: 192.100.100.95

The next free address is 192.100.100.96.

Branch 3 (10 hosts):

1. Host bits:
2^4 - 2 = 14 >= 10 → n = 4

2. Prefix:
32 - 4 = 28 → /28

3. Subnet mask: 255.255.255.240
Block size = 256 − 240 = 16

4. Starting from 192.100.100.96:
Subnet address: 192.100.100.96
First usable host: 192.100.100.97
Last usable host: 192.100.100.110
Broadcast address: 192.100.100.111

The next free address is 192.100.100.112.

WAN link (2 hosts):

1. Host bits:
2^2 - 2 = 2 >= 2 → n = 2

2. Prefix:
32 - 2 = 30 → /30

3. Subnet mask: 255.255.255.252
Block size = 256 − 252 = 4

4. Starting from 192.100.100.112:
Subnet address: 192.100.100.112
First usable host: 192.100.100.113
Last usable host: 192.100.100.114
Broadcast address: 192.100.100.115

Addresses from 192.100.100.116 to 192.100.100.255 remain unused and can be reserved for future subnets.